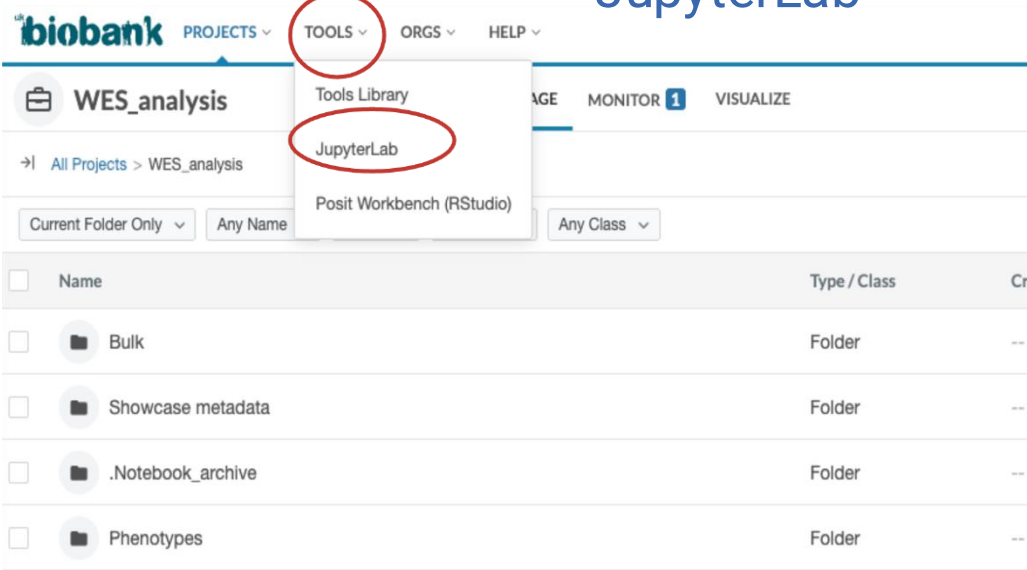


JupyterLab

a)



The screenshot shows the Biobank web interface. At the top, there is a navigation bar with 'biobank' logo, 'PROJECTS', 'TOOLS', 'ORGS', and 'HELP'. The 'TOOLS' menu is open, showing options: 'Tools Library', 'JupyterLab', and 'Posit Workbench (RStudio)'. 'JupyterLab' is circled in red. Below the navigation bar, the main content area shows 'WES_analysis' as the current project. There are filters for 'Current Folder Only' and 'Any Name'. A table lists folders: 'Bulk', 'Showcase metadata', '.Notebook_archive', and 'Phenotypes'. The table has columns for 'Name', 'Type / Class', and 'Created'.

Name	Type / Class	Created
Bulk	Folder	--
Showcase metadata	Folder	--
.Notebook_archive	Folder	--
Phenotypes	Folder	--

b) Click New JupyterLab (top right). Name your JupyterLab environment & specify the project you want to work in

c) Selecting parameters to work with JupyterLab

The screenshot shows the 'New JupyterLab' configuration window. It includes fields for Environment Name (test), Project (WES_analysis), Snapshot (Optional) (Select Snapshot), Priority (High), Cluster Configuration (Single Node selected), Instance Type (mem1_ssd1_v2_x8), Duration (in hours) (4), and Feature (PYTHON_R). At the bottom, there is an Estimated Price of £0.8912, a Cancel button, and a Start Environment button. Purple arrows point from text annotations to specific fields: 'for packages/tools not installed in RAP, snapshot allows you to load an environment with installed packages/tools' points to the Snapshot field; 'refers to time taken to load compute resources from RAP clusters>' points to the Priority field; 'each feature contains a different set of pre-installed packages that can be imported. New packages can be downloaded using !pip install <packagename>' points to the Feature field; and 'specify instance type' points to the Instance Type field.

+ New JupyterLab

Environment Name test

Project WES_analysis ...

Snapshot (Optional) Select Snapshot ...
Start the session with the same configuration and data from a previous session.

Priority ? High

Cluster Configuration Single Node Spark Cluster

Instance Type mem1_ssd1_v2_x8

Duration (in hours) 4
You can modify the duration later inside the instance.

Feature PYTHON_R

Estimated Price: £0.8912 ?
based on instance type, duration, priority, number of nodes

Cancel Start Environment

for packages/tools not installed in RAP, snapshot allows you to load an environment with installed packages/tools (refer creating a snapshot page in this document)

refers to time taken to load compute resources from RAP clusters>

Low
Request spot-instances only (The resulting analysis jobs may stay in waiting indefinitely).

Normal
Request spot-instances if available. Otherwise, request on-demand instances.

✓ **High (Default)**
Request on-demand instances for the resulting analysis jobs.

each feature contains a different set of pre-installed packages that can be imported. New packages can be downloaded using `!pip install <packagename>`

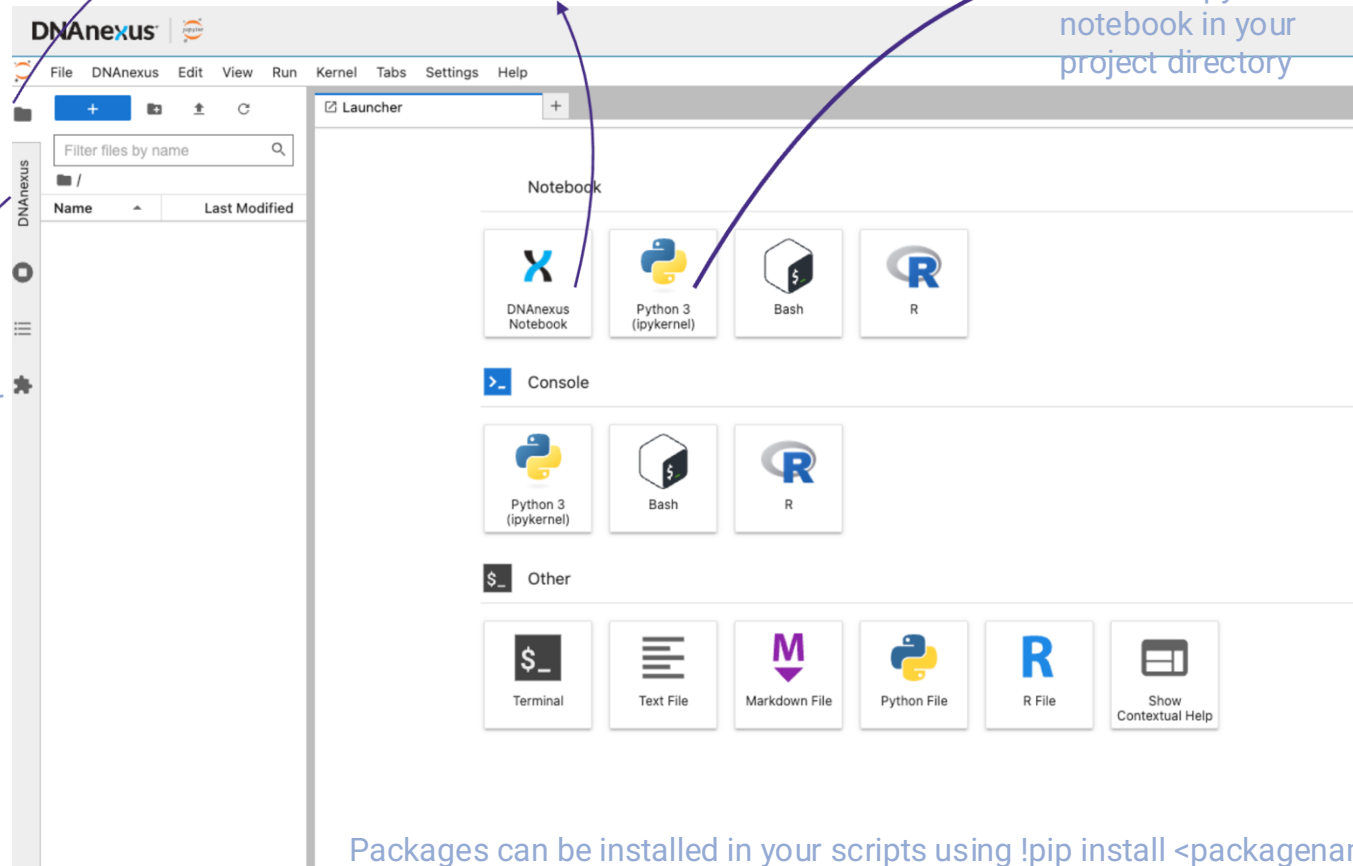
specify instance type

Files in your working environment. (Refer readme)
d) Understanding JupyterLab environment

Choosing this option **does** save the .ipynb notebook in your project directory

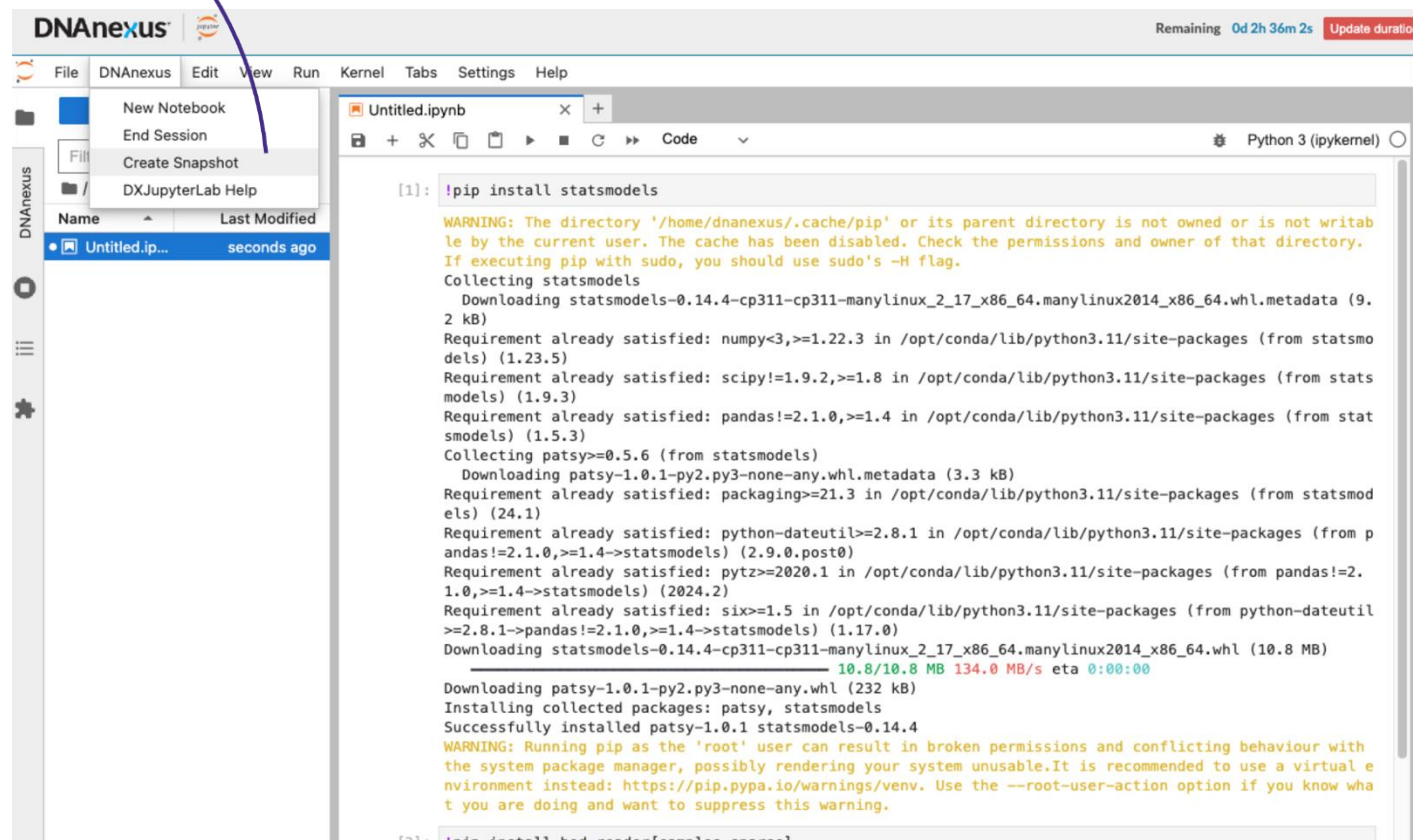
Choosing this option **does not** save the .ipynb notebook in your project directory

Files in your project directory



Packages can be installed in your scripts using `!pip install <packagename>` or
`%%bash`
`pip install <packagename>`

d) Creating a snapshot



While starting a JupyterLab environment the snapshot can be loaded from the `.Notebooks_snapshot` folder in your project